

Eliminating Quadratic Coefficient Storage

Algorithms, proofs and local verification
for counterfactual curation repair

ACL 2027 research project

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Outcome. The quadratic expansion in the original ridge-summary implementation has been removed. The stronger joint-span algorithm represents each coefficient through its current Gram/cross-moment range. Its exact-rank persistent numeric storage is $O(E_h(d + c))$, in place of $O(Kd^2 + Kdc)$. An implementation-safe envelope uses initial eligibility E_0 , because floating rank residue and sticky dense fallbacks can outlive exact rank loss.

Implemented, not just proposed. Two compact summaries, a reduced ridge decoder, deterministic projector-based basis coordinates, streaming checkpoints, and a stronger payload comparator were implemented and run here. The matrix factorization primitives are classical; the paper contribution must rest on their curation-aware guarantee and empirical consequences.

Measured result. On the fixed Civil engineering graph at $d = 768$, the original dense coefficient arrays used 234.483 MB. Joint-span arrays use 1.039 MB: a **225.6-fold reduction** (99.56%). These are owned numerical arrays, not process peak or total physical memory. A fair compact payload baseline uses only 0.256 MB, so the new summary is not a universal memory winner.

Evidence scope. The same 100 natural Civil Comments records were used, with their original labels. Curation is fixed 128-dimensional lexical hashing at threshold 0.6, independent of learner dimension. No synthetic documents or labels, semantic encoder claims, or external compute were used. These are engineering results; full NLP utility and semantic validation remain outside this local pilot.

State scope. The fast implementation repairs decoded coefficients and heads. Canonical basis coordinates remove one historical ambiguity, but sticky dense/compact mode and floating arithmetic prevent a claim of whole-state equality to fresh construction. The stronger canonical-state variant requires additional, explicitly charged re-encoding.

Measured comparisons

All cases share the same records, graph, horizon eight, regularization $\lambda = 0.01$, request generator and independent retained-data oracle. The seven-output cases use all seven original Civil toxicity annotations; they introduce no fabricated target. MB below means 10^6 bytes.

d	c	Dense	Gram factor	Joint span	Compact payload
64	1	1.699	0.119	0.071	0.022
128	1	6.641	0.271	0.172	0.043
768	1	234.483	1.628	1.039	0.256
128	7	7.249	0.879	0.183	0.047
768	7	238.133	5.278	1.047	0.260

Counts in the two new summaries are exact Python integers, charged in their separate Python live-size estimates. The table excludes those objects, identifier dictionaries, graph/corpus input, allocator slack and temporary workspace. The original dense state includes its count coordinate; that small difference is not responsible for the reduction.

The signed-Gram ablation still stores H_J densely. Its seven-output cost exposes why the joint range, rather than Gram-only compression, is needed. The compact payload comparator retains eligible FP32 features and FP64 labels, stores no permanent Gram/kernel/head, and chooses primal or dual ridge by the smaller dimension. Its persistent rows are a different storage contract.

Verification actually run

- 544 release checkpoints across three scalar-output dimension panels and two seven-output panels passed against an independently rescored oracle. The joint summary’s largest head difference was 1.11×10^{-15} ; moment difference was 4.00×10^{-15} .
- The compact decoder was actually invoked at all 544 checkpoints. Its largest oracle head difference was 1.17×10^{-15} .
- Another 100 sequential releases deleted the entire natural corpus. Joint-summary head difference was at most 2.06×10^{-14} ; the final head was exactly the prescribed zero. This is a stronger stress than the short request horizon.
- Twelve atomic-rejection checks and three order/batching comparisons passed. Twenty checkpoints per compressed method additionally compared every coefficient to the dense implementation, streamed one key at a time.
- Independent coordinate-permutation/sign audits used three actual coefficients. Streamed checkpoint roundtrip reproduced coefficients exactly, owned disjoint buffers, and supported an identical subsequent repair.

These are tolerance checks, not a certified roundoff or privacy proof. The full coefficient reference is the dense implementation; head/moment checks use a separate fresh selection-and-training oracle.

Whole-process memory and the decision it supports

Fresh processes separately constructed each method and serviced four releases. They loaded the same natural inputs and graph; they did not run oracle/audit reconstructions. Joint span used its reduced decoder. Each number is one observed high-water mark on this host, not a replicated performance estimate.

Method	Setup peak (MiB)	Final peak (MiB)
Dense coefficients	117.78	356.71
Signed Gram factors	117.74	135.42
Joint span + reduced decoder	117.78	121.28
Compact eligible payload	117.77	118.39

The final peak includes the Python/numerical runtime, original inputs, construction, requests and decoder workspace. Setup peaks must not be subtracted and interpreted as exact allocation attribution. The 225.6-fold array reduction is therefore not a 225.6-fold whole-process RAM claim.

Recommended implementation. Use joint-span coefficients for the aggregate-state method. Keep dense coefficients as an audit reference and signed-Gram factors as an explanatory ablation. Keep the compact payload method as a mandatory baseline whenever eligible-row retention is allowed. No speedup conclusion follows from these unreplicated runs.

What is solved. The summary no longer allocates one full ambient Gram per coefficient. It is built directly from bounded per-key workspaces, supports source/record substitution through the existing key algebra, and can decode a compact current coefficient without an ambient $d \times d$ matrix. The source extension here is algebraic; this new numerical run validates record deletion because the Civil preview lacks native source IDs.

What is not proved. This is not optimal physical storage, superiority to payload retention, automatic full-state canonicity, numerical exactness, privacy, or evidence of semantic NLP quality. A large corpus can still have many eligible records and substantial key metadata. Construction preflight caps coefficient arrays only; lifecycle caps must use the uniform envelope and reserve repair, decode, serialization and external input workspaces.

Reproducibility. The companion package includes all three new implementations, the exact design, per-checkpoint records, boundary results, source hashes, independent audit and process measurements. Earlier execution results remain intact. The code has no required network access during replay.

1 Removing the quadratic feature-dimension factor from persistent repair state

Scope and attribution. The dense implementation assigns a packed $d \times d$ Gram matrix to every polynomial key. This is unnecessary for ridge statistics. The previous linear-sketch lower bound permits arbitrary independent statistic coordinates; it does not establish a quadratic-in- d lower bound for rank-one ridge contributions. The matrix primitives used below—thin QR, signed spectral factorization, and low-rank update/downdate recompression—are classical, not our proposed novelty. See Matthew Brand, *Fast Low-Rank Modifications of the Thin Singular Value Decomposition*, Linear Algebra and its Applications 415(1):20–30, 2006, DOI [10.1016/j.laa.2005.07.021](https://doi.org/10.1016/j.laa.2005.07.021); the author’s institutional record is <https://www.merl.com/publications/TR2006-059>. The contribution requiring evaluation is their integration with the finite-horizon coefficient state, its source-deletion semantics, and complete memory accounting.

1.1 Intrinsic coefficient rank and exact storage

Fix the current retained collection, remaining horizon h , a frozen feature map $z_v \in \mathbb{R}^d$, and frozen responses $y_v \in \mathbb{R}^c$. For record deletion, a record’s owner is itself. For disjoint-source deletion its owner is its unique source $o(v)$, and $B(v)$ is the set of distinct blocker owners. Discard records with $o(v) \in B(v)$ and records with $|B(v)| > h$; let E_h denote the number of remaining eligible records. Each eligible record has the two possible coefficient endpoints $B(v)$ and $B(v) \cup \{o(v)\}$, with signs $+1$ and -1 , respectively. The second endpoint is absent if its degree exceeds h .

For a live key J , write its exact coefficient as

$$G_J = \sum_v a_{Jv} z_v z_v^\top, \quad H_J = \sum_v a_{Jv} z_v y_v^\top, \quad n_J = \sum_v a_{Jv}, \quad a_{Jv} \in \{-1, 0, 1\}.$$

Let $m_J = |\{v : a_{Jv} \neq 0\}|$ and let K be the number of live keys. Then $K \leq 2E_h$ and $\sum_J m_J \leq 2E_h$ before or after exact zero-key removal. Counts n_J are signed integers; only the selected-set count $n_\emptyset$ must be nonnegative.

Lemma 1.1 (Intrinsic rank envelope). *For every key,*

$$r_J := \text{rank}(G_J) \leq m_J, \quad \rho_J := \text{rank}([G_J \ H_J]) \leq m_J.$$

Consequently $\sum_J r_J \leq 2E_h$ and $\sum_J \rho_J \leq 2E_h$.

Proof. Every column of G_J and H_J lies in the span of the m_J incident feature vectors. Summing the ranks and using at most two endpoints per eligible record proves the result. The argument allows signed coefficients and repeated source owners. $\square$

The first implemented representation: signed Gram factors. Represent $G_J = F_J S_J F_J^\top$, where $F_J \in \mathbb{R}^{d \times r_J}$ and S_J is diagonal with entries $+1$ or -1 . Retain H_J densely and n_J exactly. Alternatively retain G_J in packed symmetric form, choosing whichever representation occupies fewer numeric bytes. Writing $q = d(d+1)/2$, a conservative word count is

$$\sum_J [\min\{(d+1)r_J, q\} + dc + 1] \leq 2E_h [d + dc + 2].$$

Thus persistent numeric state is $O(E_h d(c+1))$, rather than $O(Kd^2 + Kdc)$. This is an exact-arithmetic statement with rank recomputation after each collision. It does not imply better memory than every payload baseline: the constant factors, response dimension, precisions, and decoder workspace still matter. A negative or indefinite coefficient is valid and requires signed factors; replacing it by a positive-semidefinite approximation changes the query polynomial.

Theorem 1.2 (Joint-span representation). *There is an exact coefficient-only representation whose persistent numeric state uses at most*

$$\sum_J \left[d\rho_J + \frac{\rho_J(\rho_J + 1)}{2} + c\rho_J + 1 \right] \leq 2E_h \left(\frac{3}{2}d + c + \frac{3}{2} \right)$$

real/integer words. With key indices and alive public identifiers, logical storage is $O(E_h(d+c+h)+U)$ words, where U is the number of alive public deletion units. This representation supports the same finite-horizon record and disjoint-source deletion polynomial without retained feature rows, response rows, or the original blocker graph.

Proof. Choose an orthonormal basis $U_J \in \mathbb{R}^{d \times \rho_J}$ of $\text{range}([G_J \ H_J])$, and store

$$A_J = U_J^\top G_J U_J, \quad B_J = U_J^\top H_J, \quad n_J.$$

Because G_J is symmetric, its row and column spaces lie in the same subspace. Hence $G_J = U_J A_J U_J^\top$ and $H_J = U_J B_J$ exactly. The displayed count stores U_J , the packed symmetric A_J , B_J , and the count. Use $\rho_J \leq d$, $\sum_J \rho_J \leq 2E_h$, and $K \leq 2E_h$ for the upper bound. Each key has at most h identifier incidences; its dictionaries and degree/inverse indices therefore cost $O(hE_h + E_h)$ logical words. No record-level decomposition of G_J or H_J is retained. $\square$

Why the joint range is necessary. The range of G_J alone is insufficient. Two signed contributions with the same nonzero feature z , opposite signs, and distinct labels can give $G_J = 0$ but $H_J = z(y_1 - y_2)^\top \neq 0$. A scheme that stores cross moments only in the Gram range silently loses information in this case. The joint-span representation covers it without storing the original labels.

1.2 Repairs, canonicity, and work

The polynomial-key update is unchanged: substitute deleted variables by one, merge colliding keys, and remove keys of degree above the decreased horizon. For a joint-span collision, let Q span $[U_1 \ U_2]$, set $P_i = Q^\top U_i$, and form

$$A = P_1 A_1 P_1^\top + P_2 A_2 P_2^\top, \quad B = P_1 B_1 + P_2 B_2, \quad n = n_1 + n_2.$$

If V spans $\text{range}([A \ B])$, replace the state by $(QV, V^\top AV, V^\top B, n)$ and discard its predecessor representations. The signed-Gram variant instead obtains a thin QR factorization $[F_1 \ F_2] = QR$, diagonalizes $R \text{diag}(S_1, S_2) R^\top$, and absorbs square roots of absolute eigenvalues into the new factor. Exact zero eigenvalues can be discarded; negative eigenvalues cannot.

Proposition 1.3 (Exact repair and state scope). *In exact arithmetic these operations decode to the canonical coefficient map of the retained dataset at its remaining horizon. If each factorization uses a deterministic convention depending only on its decoded current coefficient, the encoded state is also a function of that current map and the public membership/horizon state. It does not retain an original basis.*

Proof. The factor operations preserve the sum of the decoded coefficients exactly. Substitution and decreased-horizon truncation therefore commute with decoding. The finite-horizon polynomial identity identifies the result with fresh retained-data construction. A deterministic encoding of that map preserves this state identity. $\square$

For example, a canonical joint basis can be obtained by processing the columns of $[G_J H_J]$ in fixed order, orthogonalizing each against earlier accepted columns, skipping exactly zero residuals, and fixing normalization signs. These columns can be generated from a small current factor, so an entire dense ambient matrix need not persist. For spectral encoding, order distinct eigenvalues, then canonically orient each eigenspace by applying its projector to the coordinate vectors in fixed order and orthogonalizing. An arbitrary QR or eigensolver output is not automatically this canonical state, especially for repeated eigenvalues. The numerical implementation must therefore claim coefficient/head fidelity, not bitwise canonical factor equality.

For colliding widths p, q , set $s = p + q \leq 2d$. A streamed factor collision uses $O(ds + s^2 + dc)$ additional numeric workspace and, conservatively, $O(ds^2 + s^3 + s^2c)$ arithmetic. The persistent-state bounds do not promise the old dense algorithm’s $O(d^2)$ cost per collision: repeated recompression can be slower. If C collisions occur, charge the sum of these actual per-collision bounds, in addition to the unchanged key-index work. Process collisions one at a time rather than allocating a dense matrix for every key. A common dense ridge decoder separately uses $O(d^2 + dc)$ workspace and typically $O(d^3 + d^2c)$ factor/solve work. Construction must aggregate directly into factors or bounded per-key workspaces; constructing the old K -matrix state and compressing afterward does not cure its construction peak. Checkpoint export and array copies must also be charged.

1.3 What the floating-point implementation actually guarantees

Exact rank and finite-precision rank are different. Mathematical cancellation may leave tiny nonzero FP64 eigenvalues, while thresholding them would introduce an additional approximation. Retaining every numerically nonzero eigenvalue avoids a user-selected truncation tolerance, but does not make factorization exact or make stored width equal to mathematical rank. QR, eigendecomposition, reconstruction, and summation still introduce roundoff. The current- E_h bounds above belong to the exact-rank theorem, not automatically to software that keeps numerical residue or leaves formerly dense entries dense.

Proposition 1.4 (Implementation-safe initial-state envelope). *Consider the signed-Gram implementation with these explicit rules: (i) a factor-factor collision returns at most the sum of the input widths; (ii) dense storage is chosen only if its numeric bytes do not exceed the candidate factor bytes; (iii) a collision involving a dense Gram remains dense; and (iv) updates only rename, merge, or prune coefficient keys. Then persistent Gram-array bytes, and also the total Gram/cross/count-array bytes, never increase during a valid deletion sequence. For FP64 factors, int8 signs, FP64 cross moments, and int64 counts, streamed rank-one construction with E_0 initially eligible records gives the envelope*

$$S_{\text{numeric}}(t) \leq S_{\text{numeric}}(0) \leq 2E_0(8d + 1 + 8dc + 8) \quad \text{bytes.}$$

This bound excludes Python objects, key metadata, allocator slack, serialization duplication, and transient repair/decoder workspace.

Proof. A factor of width r occupies $8dr + r$ bytes. A factor-factor collision has width at most the sum of its input widths, so its factor bytes are at most their sum; choosing a smaller packed representation preserves this bound. If an input is dense, retaining one packed matrix costs no more than the existing dense input. Cross arrays and counts need one copy per resulting key, and key count cannot increase. Each original eligible record inserts at most two rank-one endpoint contributions, proving the initial bound. No argument here depends on exact spectral cancellation. In particular, the statement uses E_0 , not the smaller current E_h . $\square$

A distinct safe envelope for joint spans. Joint-span numeric bytes need not be monotone: the small core has a quadratic term, so merging two bases can increase core bytes even when

their total width does not increase. For a joint implementation, assign each initial endpoint one abstract ancestry token. Tokens are used only for the analysis; they need not identify or retain a payload. Merge token sets on collisions and discard them on pruning. Current entries have disjoint ancestry and use at most $2E_0$ tokens altogether. A factor entry of width ρ uses at most its token count a , and $\rho \leq d$. Its packed-core storage therefore satisfies

$$d\rho + \rho(\rho + 1)/2 + c\rho + 1 \leq (3d/2 + c + 1/2)a + 1.$$

If a dense entry is created only when its byte count does not exceed the candidate factor byte count, charge it to that same token bound. Keeping it dense on later collisions cannot exceed its earlier charge, and its token set only grows. Hence the persistent numeric envelope is $2E_0(3d/2 + c + 3/2)$ words with packed cores. If software stores the entire $\rho \times \rho$ core instead, use the larger valid envelope $2E_0(2d + c + 1)$ words. This is a bound on all future numeric allocations in the persistent state, not on per-collision temporary allocation. An initial measured joint-state size is not by itself a safe future memory cap.

The numeric envelope is not a full physical-memory or privacy theorem. Persistent factors can encode recoverable record information whenever the underlying coefficient does; hiding rows syntactically is not a privacy guarantee. No predecessor factors, old basis, or payload snapshots may be retained under the stated current-summary contract. Allocator remnants, backups, and previously released models remain outside that abstract-state contract. Whole-process peak RSS, counted owned arrays, serialized bytes, and common decoder workspace should be reported separately.

Limits of a single shared span. A global basis $U \in \mathbb{R}^{d \times r}$ shared by all keys gives storage $O(dr + K(r^2 + rc))$. It is useful when $r \ll d$, as can happen when a tiny pilot has fewer examples than feature dimensions. When a realistic corpus spans all d features, $r = d$ and this does not remove the Kd^2 term. Moreover a basis frozen from the original corpus can retain deleted-only directions. The per-key rank envelope above neither assumes global low rank nor preserves such an original basis.

Claims supported by this addition. The exact algorithm eliminates the unavoidable-looking quadratic feature factor from persistent ridge-summary storage, with $O(E_h(d + c))$ words for joint spans. The implemented signed-Gram variant has an auditable $O(E_0d(c + 1))$ numeric envelope even without numerical rank truncation. Neither statement establishes universal superiority to an eligible-payload baseline, smaller whole-process peak memory, faster repairs, bitwise canonical floating-point state, or a privacy guarantee. Those distinctions are part of the algorithm specification, not qualifications to omit from the empirical comparison.

1.4 Canonical basis orientation and compact decoding

The implemented projector-pivot convention. For an orthonormal $U \in \mathbb{R}^{d \times \rho}$, initialize $V = U^\top \in \mathbb{R}^{\rho \times d}$. Repeatedly select the first column attaining the largest residual squared norm, normalize that column to o_j , and subtract its orthogonal projection from every column of V . After ρ steps let $O = [o_1 \ \cdots \ o_\rho]$. The reoriented state is

$$U_{\text{can}} = UO, \quad A_{\text{can}} = O^\top AO, \quad B_{\text{can}} = O^\top B.$$

This is pivoted Gram-Schmidt on projected coordinate vectors, performed in small basis coordinates. It need not form the ambient projector $UU^\top$. In exact arithmetic there is a positive pivot at every step before ρ , since $U^\top$ has row rank ρ .

Proposition 1.5 (Rotation invariance of the basis convention). *Suppose two orthonormal bases represent the same subspace and coefficient: $\hat{U} = UR$, $\hat{A} = R^\top AR$, and $\hat{B} = R^\top B$, with R orthogonal. The projector-pivot convention produces identical $(U_{\text{can}}, A_{\text{can}}, B_{\text{can}})$ in exact arithmetic. It depends on the current subspace and coefficient, not on the original construction basis.*

Proof. Initially $\hat{V} = R^\top V$. Orthogonal transformations preserve all residual column norms, so both procedures select the same pivot, including the first-index tie rule. Its normalized vector satisfies $\hat{o}_j = R^\top o_j$. Orthogonal projection subtraction preserves $\hat{V} = R^\top V$ at the next step. Induction gives $\hat{O} = R^\top O$. Consequently $\hat{U}\hat{O} = UO$, and substitution into the two small-core transformations proves equality. $\square$

This proposition is conditional on the represented subspace. If numerical residue keeps extra basis directions, two repair histories can represent different approximate subspaces. Even on the same mathematical subspace, roundoff and near-tied pivots can produce different numerical choices. Thus the implemented orientation is a deterministic convention, not a proof of bitwise canonical floating-point state or exact numerical rank. It also does not justify retaining a larger original basis indefinitely.

Basis orientation does not canonicalize representation mode. The fast implementation can keep a coefficient dense after an earlier collision, even when a fresh construction would choose a compact factor. Likewise, an initial contributor upper bound can affect a construction-mode decision. These choices can depend on repair history even in exact arithmetic. Canonical basis orientation removes basis rotations within a chosen subspace; it does not remove this dense-versus-compact mode history. The fast implementation consequently claims decoded coefficient/head equivalence, not literal canonical-state equality. To instantiate the stronger encoded-state theorem, the mode, width, and basis must all be chosen from the actual current coefficient, for example by recompressing dense entries on collisions or performing a canonical export. Such a step can require $O(d^3)$ work per dense entry and its temporary memory must be charged. It is not a free consequence of the pivot-orientation routine.

Proposition 1.6 (Compact current-head decoder). *For a nonempty selected set with exact count $n > 0$, suppose its current moments are $G = UAU^\top \succeq 0$ and $H = UB$, with $U^\top U = I_\rho$. For averaged ridge regularization $\lambda > 0$, the unique fitted head is*

$$W^* = (G + \lambda n I_d)^{-1} H = U(A + \lambda n I_\rho)^{-1} B.$$

Given the stored orthonormal joint representation, this requires $O(\rho^3 + \rho^2 c + d\rho c)$ arithmetic and $O(\rho^2 + \rho c + dc)$ additional numeric workspace. There is no additional $O(d\rho^2)$ basis-construction charge at decode time. If the state uses a dense fallback, the corresponding solve has $O(d^2 + dc)$ workspace and $O(d^3 + d^2 c)$ arithmetic.

Proof. The right-hand side is in $\text{range}(U)$. The orthogonal complement has zero right-hand side and positive regularization, so its solution component is zero. Write $W = UZ$ and multiply the normal equation by $U^\top$ to obtain $(A + \lambda n I_\rho)Z = B$. A Cholesky factorization, small solves, and multiplication by U give the stated costs. The positive-semidefinite assumption applies to the current selected Gram, not to every signed polynomial coefficient. $\square$

The cases $n = 0$ and $\rho = 0$ use the previously specified zero-head convention where applicable; expressions divided by n below require $n > 0$. In floating-point arithmetic, approximate orthogonality can make the compact equation differ slightly from the ambient normal equation. An ambient residual can still be evaluated without materializing G : $UA(U^\top \hat{W}) + \lambda n \hat{W} - UB$.

Proposition 1.7 (Head error from moment error and solve residual). *Let (G_*, H_*, n) be the true retained moments, where $G_* \succeq 0$ and $n > 0$ is exact. Let $(\tilde{G}, \tilde{H}, n)$ be represented approximate moments and let $\tilde{W}$ be any computed head. Suppose*

$$\|\tilde{G} - G_*\|_{\text{op}} \leq \varepsilon_G, \quad \|\tilde{H} - H_*\|_F \leq \varepsilon_H, \quad \|(\tilde{G} + \lambda n I)\tilde{W} - \tilde{H}\|_F \leq \rho_{\text{res}}.$$

Then, with $W_ = (G_* + \lambda n I)^{-1} H_*$,*

$$\|\tilde{W} - W_*\|_F \leq \frac{\rho_{\text{res}} + \varepsilon_H + \varepsilon_G \|\tilde{W}\|_F}{\lambda n}.$$

Proof. The true normal-equation residual equals the represented residual plus $(G_* - \tilde{G})\tilde{W} + (\tilde{H} - H_*)$. Apply the triangle inequality and $\|(G_* + \lambda n I)^{-1}\|_{\text{op}} \leq 1/(\lambda n)$. No positive-semidefinite assumption on $\tilde{G}$ is needed for this error bound, although an actual solver still must return a finite head. $\square$

A small residual alone does not bound moment approximation error. The displayed inequality becomes a numerical certificate only if all three input bounds include rigorous floating-point error allowances. Measured differences against a fresh oracle and ordinary floating-point residuals are diagnostic validation, not such a certificate. The exact count assumption is essential; a count error also perturbs the regularization term and cannot be silently omitted.